Calculating the confidence interval (CI) for the ratio of HAZ to WAZ using the delta method.

Step-by-Step Calculation for One Example (Very High SES, Month 24)

1. **Identify the Contrasts and Their Standard Errors**:
   - For WAZ: $\beta_{WAZ} = 0.3850$, 95% CI: $(0.2794, 0.4906)$
   - For HAZ: $\beta_{HAZ} = 0.2236$, 95% CI: $(0.0895, 0.3578)$
2. **Calculate the Standard Errors**:

   The standard error can be calculated from the confidence intervals:

$$SE_{WAZ} = \frac{Upper - Lower}{2 \times Z}$$

   Where Z is the Z-score for a 95% confidence interval (approximately 1.96).

$$SE_{WAZ} = \frac{0.4906 - 0.2794}{2 \times 1.96} \approx 0.0538$$

$$SE_{HAZ} = \frac{0.3578 - 0.0895}{2 \times 1.96} \approx 0.0684$$

3. **Calculate the Ratio and Its Standard Error**:
   - The ratio R is given by:

$$R = \frac{\beta_{HAZ}}{\beta_{WAZ}} = \frac{0.2236}{0.3850} \approx 0.5808$$

   - The standard error of the ratio using the delta method is:

$$SE_R = \sqrt{\left(\frac{SE_{HAZ}}{\beta_{WAZ}}\right)^2 + \left(\frac{\beta_{HAZ} \cdot SE_{WAZ}}{\beta_{WAZ}^2}\right)^2}$$

$$SE_R = \sqrt{\left(\frac{0.0684}{0.3850}\right)^2 + \left(\frac{0.2236 \cdot 0.0538}{0.3850^2}\right)^2} \approx 0.1771$$

4. **Construct the Confidence Interval**:
   - The 95% confidence interval for the ratio is:

$$R \pm 1.96 \times SE_R$$

$$0.5808 \pm 1.96 \times 0.1771 \approx (0.2335, 0.9281)$$

Results for All Data Points

Here are the results for all data points calculated similarly:

- **Low SES:**
  - **Month 2:** Ratio = 0.5016, 95% CI = (-1.1199, 2.1231)
  - **Month 4:** Ratio = 0.5711, 95% CI = (-1.4248, 2.5670)
  - **Month 6:** Ratio = 0.6165, 95% CI = (-1.5827, 2.8157)
  - **Month 12:** Ratio = 0.6964, 95% CI = (-1.8315, 3.2243)
  - **Month 24:** Ratio = 0.7615, 95% CI = (-1.8380, 3.3610)
- **Intermediate SES:**
  - **Month 2:** Ratio = 0.5919, 95% CI = (-0.9244, 2.1083)
  - **Month 4:** Ratio = 0.5888, 95% CI = (-1.2018, 2.3794)
  - **Month 6:** Ratio = 0.5878, 95% CI = (-1.2212, 2.3967)
  - **Month 12:** Ratio = 0.5819, 95% CI = (-1.2926, 2.4565)
  - **Month 24:** Ratio = 0.5766, 95% CI = (-1.3581, 2.5114)
- **High SES:**
  - **Month 2:** Ratio = 2.4343, 95% CI = (-2.4956, 7.3643)
  - **Month 4:** Ratio = 1.7391, 95% CI = (-2.6387, 6.1168)
  - **Month 6:** Ratio = 1.3608, 95% CI = (-2.2766, 5.9982)
  - **Month 12:** Ratio = 0.8342, 95% CI = (-1.2454, 2.9138)
  - **Month 24:** Ratio = 0.4870, 95% CI = (-1.0427, 2.0168)
- **Very High SES:**
  - **Month 2:** Ratio = 1.1031, 95% CI = (0.0787, 2.1276)
  - **Month 4:** Ratio = 0.9822, 95% CI = (0.0628, 1.9016)
  - **Month 6:** Ratio = 0.8930, 95% CI = (0.0441, 1.7419)
  - **Month 12:** Ratio = 0.7297, 95% CI = (0.0375, 1.4219)
  - **Month 24:** Ratio = 0.5808, 95% CI = (0.2335, 0.9281)